



SYDNEY BOYS HIGH SCHOOL

MOORE PARK, SURRY HILLS

2004

YEAR 12

**HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK # 3**

Mathematics Extension 1

General Instructions

- Working time – 90 minutes.
- Reading Time – 5 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may not be awarded for messy or badly arranged work

Total Marks - 66

- Attempt *all* questions
- *All* questions are of equal value
- Return your answers in 3 booklets, one for each section. Each booklet must show your student number.

Examiner: *Mr R Dowdell*

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left\{ x + \sqrt{x^2 - a^2} \right\}, |x| > |a|$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left\{ x + \sqrt{x^2 + a^2} \right\}$$

NOTE: $\ln x = \log_e x$

Section A:**Question 1: (11 marks)**

Marks

- (a) Evaluate $\int_0^2 \frac{dx}{\sqrt{16-x^2}}$ 2
- (b) Evaluate
- (i) $\lim_{x \rightarrow 0} \frac{\sin 3x}{4x}$ 3
- (ii) $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 7x}$
- (c) Use the substitution $u = \ln x$ to find $\int \frac{dx}{x\sqrt{1-(\ln x)^2}}$. 2
- (d) Differentiate $\log_e(\sin^3 x)$, writing your answer in simplest form. 2
- (e) Differentiate with respect to x , $(\tan^{-1} x)^2$. 2

Question 2: (11 marks)

Marks

(a) (i) Write down the domain and range of $y = \sin^{-1}(\sin x)$.

(ii) Draw a neat sketch of $y = \sin^{-1}(\sin x)$.

3

(b) Given that $y = \sin^{-1}(\sqrt{x})$, show that $\frac{dy}{dx} = \frac{1}{\sin 2y}$.

3

(c) Show that the derivative of $x \tan x - \ln(\sec x)$ is $x \sec^2 x$.

Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} x \sec^2 x \, dx$.

3

(d) If $y = 10^x$, find $\frac{dy}{dx}$ when $x = 1$.

2

Section B:

Question 3: (11 marks) **START A NEW BOOKLET**

Marks

- (a) Consider the function $y = 4 \sin\left(x + \frac{\pi}{6}\right)$, $\frac{\pi}{3} \leq x \leq \frac{4\pi}{3}$.
- (i) Find the inverse function of y , and write down its domain. 4
 - (ii) Sketch the inverse function of y .
- (b) (i) On the same axes, draw the graphs of $y = \tan^{-1} x$ and $y = \cos^{-1} x$, showing the important features. Mark the point P where the curves intersect. 5
- (ii) Show that, if $\tan^{-1} x = \cos^{-1} x$, then $x^4 + x^2 - 1 = 0$. Hence, find the coordinates of P , correct to 2 decimal places.
- (c) Show that $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$ 2

Question 4: (11 marks)

Marks

- (a) (i) Draw a neat sketch of $y = \cos^{-1} x$. State its domain and range.
 (ii) Shade the area bounded by $y = \cos^{-1} x$ and the x and y axes on your diagram. 4
 (iii) Calculate the area of the region specified in (ii).
- (b) Differentiate $y = \log_e \left(\frac{2x}{(x-1)^2} \right)$. Write your answer in simplest form. 2
- (c) The rate of change of temperature T° , of an object is given by the equation $\frac{dT}{dt} = k(T - 16)$ degrees per minute, k a constant.
 (i) Show that the function $T = 16 + Pe^{kt}$, where P is a constant and t the time in minutes, satisfies the equation.
 (ii) If initially $T = 0$ and after 10 minutes $T = 12$, find the values of P and k . 5
 (iii) Find the temperature of the object after 15 minutes.
 (iv) Sketch the graph of T as a function of t and describe its behaviour as t continues to increase.

Section C:

Question 5: (11 marks) **START A NEW BOOKLET**

Marks

- (a) It is known that $\ln x + \sin x = 0$ has a root close to $x = 0.5$. Use one application of Newton's method to obtain a better approximation (to 2 decimal places). 2
- (b) The acceleration of a particle P is given by the equation $\ddot{x} = 8x(x^2 + 1) \text{ ms}^{-2}$, where x is the displacement of P from the origin in metres after t seconds, with movement being in a straight line.
- Initially the particle is projected from the origin with a velocity of 2 ms^{-1} .
- (i) Show that the velocity of the particle can be expressed as $v = 2(x^2 + 1)$. 6
- (ii) Hence, show that the equation describing the displacement of the particle at time t is given by $x = \tan 2t$.
- (iii) Determine the velocity of the particle at time $\frac{\pi}{8}$ seconds.
- (c) The arc of the curve $y = \sin^{-1} x$ between $x = 0$ and $x = 1$ is rotated about the x axis. Use Simpson's Rule with three function values to estimate the volume of the solid formed. 3

Question 6: (11 marks)

Marks

- (a) The velocity $v \text{ ms}^{-2}$ of a particle moving in simple harmonic motion along the x axis is given by the expression $v^2 = 28 + 24x - 4x^2$.

- (i) Between which two points is the particle oscillating?
- (ii) What is the amplitude of the motion?
- (iii) Find the acceleration in terms of x .
- (iv) Find the period of the oscillation.
- (v) If the particle starts from the point furthest to the right, find the displacement in terms of t .

6

- (b) A stone is thrown from the top of a vertical cliff over the water of a lake. The height of the cliff is 8 metres above the level of the water, the initial speed of the stone is 10 ms^{-1} and the angle of projection is $\theta = \tan^{-1}\left(\frac{3}{4}\right)$ above the horizontal.

The equations of motion of the stone, with air resistance neglected, are $\ddot{x} = 0$ and $\ddot{y} = -g$.

- (i) By taking the origin O as the base of the cliff, show that the horizontal and vertical components of the stone's displacement from the origin after t seconds are given by $x = 8t$ and

5

$$y = -\frac{1}{2}gt^2 + 6t + 8.$$

- (ii) Hence, or otherwise, calculate the time which elapses before the stone hits the lake and find the horizontal distance of the point of contact from the base of the cliff. (Assume $g = 10 \text{ ms}^{-2}$.)

End of Paper



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Mathematics Extension 1

Sample Solutions

SECTION	MARKER
A	Ms Opferkuch
B	Ms Nesbitt
C	Mr Bigelow

Section A

Question 1

$$\begin{aligned}
 \text{(a)} \quad \int_0^2 \frac{1}{\sqrt{16-x^2}} dx &= \int_0^2 \frac{1}{\sqrt{4^2-x^2}} dx \\
 &= \left[\sin^{-1} \frac{x}{4} \right]_0^2 \\
 &= \sin^{-1} \frac{1}{2} \\
 &= \frac{\pi}{6}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) (i)} \quad \lim_{x \rightarrow \infty} \frac{\sin 3x}{4x} &= \frac{3}{4} \lim_{x \rightarrow \infty} \frac{\sin 3x}{3x} \\
 &= \frac{3}{4} \times 1 \\
 &= \frac{3}{4}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \lim_{x \rightarrow \infty} \frac{\sin 3x}{\sin 7x} &= \lim_{x \rightarrow \infty} \frac{\sin 3x}{3x} \times \frac{7x}{\sin 7x} \\
 &= \frac{3}{7} \lim_{x \rightarrow \infty} \frac{\sin 3x}{3x} \times \frac{7x}{\sin 7x} \\
 &= \frac{3}{7} \times 1 \\
 &= \frac{3}{7}
 \end{aligned}$$

$$\text{(c)} \quad \int \frac{dx}{x\sqrt{1-(\ln x)^2}}$$

$$\text{Let } u = \ln x$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$du = \frac{1}{x} dx$$

$$\begin{aligned}
 \int \frac{dx}{x\sqrt{1-(\ln x)^2}} &= \int \frac{1}{\sqrt{1-u^2}} du \\
 &= \sin^{-1} u + C \\
 &= \sin^{-1}(\ln x) + C
 \end{aligned}$$

$$\text{(d)} \quad \log_e(\sin^3 x)$$

$$\text{Let } u = \sin^3 x$$

$$\frac{du}{dx} = 3\sin^2 x \cos x$$

$$\text{Let } y = \log_e u$$

$$\frac{dy}{du} = \frac{1}{u}$$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\
 &= \frac{1}{u} \times 3\sin^2 x \cos x \\
 &= \frac{1}{\sin^3 x} \times 3\sin^2 x \cos x \\
 &= \frac{3\cos x}{\sin x} \\
 \therefore \frac{dy}{dx} &= 3\cot x
 \end{aligned}$$

(e) $\frac{d}{dx}(\tan^{-1} x)^2$

Let $u = \tan^{-1} x$

$$\frac{du}{dx} = \frac{1}{1+x^2}$$

Let $y = u^2$

$$\frac{dy}{du} = 2u$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= 2u \times \frac{1}{1+x^2}\end{aligned}$$

$$\therefore \frac{dy}{dx} = \frac{2 \tan^{-1} x}{1+x^2}$$

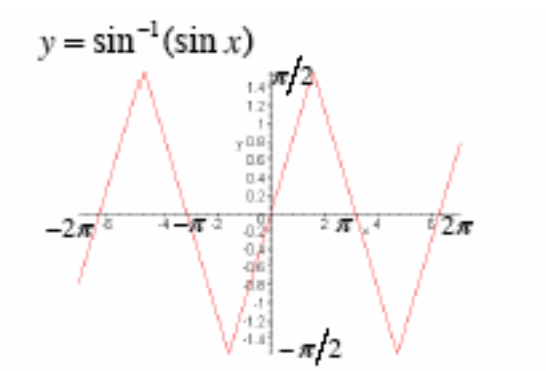
Question 2

(a) (i) $y = \sin^{-1}(\sin x)$

Domain $\{x : x \in \mathbb{R}\}$

Range $\left\{y : -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}\right\}$

(ii)



(b) $y = \sin^{-1}(\sqrt{x})$

$$\sin y = \sqrt{x}$$

$$\sin^2 y = x$$

$$\therefore x = \sin^2 y$$

$$\frac{dx}{dy} = 2 \sin y \cos y$$

$$= \sin 2y$$

$$\therefore \frac{dy}{dx} = \frac{1}{\sin 2y}$$

(c) (i) $y = x \tan x - \ln(\sec x)$

Now $\frac{d}{dx} x \tan x$

Let $u = x \quad v = \tan x$

$$\frac{du}{dx} = 1 \quad \frac{dv}{dx} = \sec^2 x$$

$$\begin{aligned} \therefore \frac{d}{dx} (x \tan x) &= u \frac{dv}{dx} + v \frac{du}{dx} \\ &= (x)(\sec^2 x) + (\tan x)(1) \\ &= x \sec^2 x + \tan x \end{aligned}$$

Now $\frac{d}{dx} \ln(\sec x)$

Let $u = \sec x$

$y = \ln u$

$= (\cos^{-1} x)$

$$\frac{dy}{du} = \frac{1}{u}$$

$$\frac{du}{dx} = -(\cos x)^{-2} (-\sin x)$$

$$= \frac{\sin x}{\cos^2 x}$$

$$= \tan x \sec x$$

$$\therefore \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{1}{u} \times \tan x \sec x$$

$$= \frac{1}{\sec x} \times \tan x \sec x$$

$$= \tan x$$

$$\therefore y = x \tan x - \ln(\sec x)$$

$$\frac{dy}{dx} = x \sec^2 x + \tan x - \tan x$$

$$= x \sec^2 x$$

(c)

$$\begin{aligned} \text{(ii)} \quad \int x \sec^2 x \, dx &= \left[x \tan x - \ln(\sec x) \right]_0^{\frac{\pi}{4}} \\ &= \left\{ \frac{\pi}{4} \tan \frac{\pi}{4} - \ln(\sec \frac{\pi}{4}) \right\} - \{ 0 \tan 0 - \ln(\sec 0) \} \\ &= \left\{ \frac{\pi}{4} (1) - \ln(\sqrt{2}) \right\} - \{ -\ln(1) \} \\ &= \frac{\pi}{4} - \ln \sqrt{2} \\ &= \frac{\pi}{4} - \frac{1}{2} \ln 2 \\ &= \frac{\pi - 2 \ln 2}{4} \end{aligned}$$

$$\text{(d)} \quad y = 10^x$$

$$\log_{10} y = \log_{10} 10^x$$

$$\log_{10} y = x \log_{10} 10$$

$$x = \log_{10} y$$

$$x = \frac{\log_e y}{\log_e 10}$$

$$x = \frac{1}{\log_e 10} \times \log_e y$$

$$x \log_e 10 = \log_e y$$

$$\therefore y = e^{x \log_e 10}$$

$$\therefore \frac{dy}{dx} = \log_e 10 \times e^{x \log_e 10}$$

when $x = 1$

$$\frac{dy}{dx} = \log_e 10 \times e^{(1) \log_e 10}$$

$$= \log_e 10 \times 10$$

$$= 10 \log_e 10$$

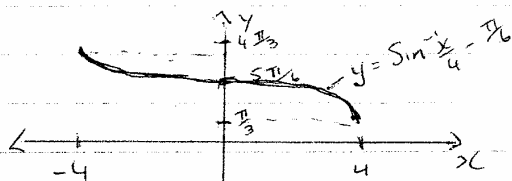
QUESTION 3

(a) Inverse $x = 4 \sin(y + \frac{\pi}{6})$ $\frac{\pi}{3} \leq y \leq \frac{4\pi}{3}$ a(i)

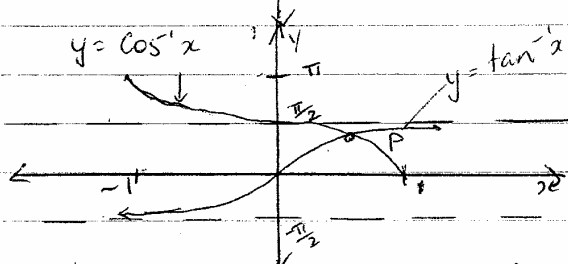
(i) $y + \frac{\pi}{6} = \sin^{-1} \frac{x}{4}$

$y = \sin^{-1} \frac{x}{4} - \frac{\pi}{6}$

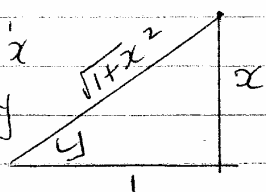
Domain $-4 \leq x \leq 4$



b)(i)



(ii) $\tan^{-1} x = y = \cos^{-1} x$
 $\tan y = x = \cos y$



$x = \frac{1}{\sqrt{1+x^2}}$

$x^2(1+x^2) = 1$

$x^4 + x^2 - 1 = 0$

$x^2 = \frac{-1 \pm \sqrt{1+4}}{2}$

$x^2 = 0.618$ ($x^2 > 0$)

$x = 0.79$, $y = 0.67$ (2 d.p.)

(c) $x = \tan^{-1}(\frac{1}{4})$ $y = \tan^{-1}(\frac{3}{5})$

$\tan x = \frac{1}{4}$ $\tan y = \frac{3}{5}$

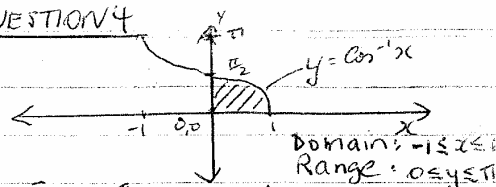
$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$

$\tan(x+y) = \frac{\frac{1}{4} + \frac{3}{5}}{1 - \frac{1}{4} \cdot \frac{3}{5}} = 1$

$x + y = \frac{\pi}{4}$

$\tan^{-1}(\frac{1}{4}) + \tan^{-1}(\frac{3}{5}) = \frac{\pi}{4}$

QUESTION 4



$A = \int_0^{\pi/2} \cos y \, dy$

$= [\sin y]_0^{\pi/2}$

$= 1 - 0 = 1 \text{ u}^2$

(b) $y = \log_e 2x - 2 \log_e(x-1)$

$= \frac{1}{x} - \frac{2}{x-1}$ or $-\frac{x+1}{x(x-1)}$

(c) $\frac{dT}{dt} = k(T-16)$

(i) $\frac{dT}{dt} = \frac{1}{k(T-16)}$

$t = \frac{1}{k} \log_e(T-16) + C$

$k(t-C) = \log_e(T-16)$

$T-16 = e^{kt-kC}$

$T-16 = Pe^{kt}$ ($P = e^{-kC} = \text{constant}$)

$T = 16 + Pe^{kt}$ as required

(ii) $T = 16 + Pe^{kt}$

$T=0, t=0$ $P = -16$

$T = 16 - 16e^{kt}$

$T=12, 12 = 16 - 16e^{kt}$

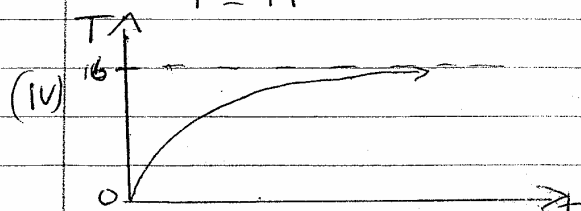
$-4 = -16e^{10k}$

$\frac{1}{4} = e^{10k}$, $k = \frac{1}{10} \log_e \frac{1}{4}$

$k \approx -0.1326$

(iii) $t=15$ $T = 16 - 16e^{-0.1326 \times 15}$

$T = 14^\circ$



as $t \rightarrow \infty$, $16e^{kt} \rightarrow 0$

$T \rightarrow 16$

QUESTION 5.

(a). Let $f(x) = \ln x + \sin x$.
 $f'(x) = \frac{1}{x} + \cos x$. (1/2 for diff)

Qf $x_1 = 0.5$.
 then $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$
 $= 0.5 - \frac{\ln 0.5 + \sin 0.5}{\frac{1}{0.5} + \cos 0.5}$ $\downarrow$ (1) NB
 $= 0.5 - -0.07427$ $\downarrow$ if calculator
 $= \boxed{0.57}$ (2 D.P.). $\boxed{2}$ is in degree
 made 0.73 (4 MARK)

(b)(i). $\ddot{x} = 8x(x^2 + 1)$ and when $t=0$, $x=0$, $v=2$.

$\frac{d}{dt}(\frac{1}{2}v^2) = 8x^3 + 8x$

$\frac{1}{2}v^2 = 2x^4 + 4x^2 + c$

now $v=0$ when $x=0$

$\therefore \frac{1}{2}v^2 = 0 + 0 + c$

$c = 2$

$\therefore \frac{1}{2}v^2 = 2x^4 + 4x^2 + 2$

$v^2 = 4x^4 + 8x^2 + 4$

$v^2 = 4(x^4 + 2x^2 + 1)$

$= 4(x^2 + 1)^2$

$v = \pm 2(x^2 + 1)$ (now $v=0$ when $x=0$ $\therefore v \neq -2(x^2 + 1)$)

$\therefore \boxed{v = 2(x^2 + 1)}$ $\boxed{2}$

must show this.
 $\downarrow$

(ii) $\frac{dx}{dt} = 2(x^2 + 1)$

$\frac{dt}{dx} = \frac{1}{2(x^2 + 1)}$

$t = \frac{1}{2} \tan^{-1} x + c$

now $t=0$, when $x=0$.

$\therefore 0 = \frac{1}{2} \tan^{-1} 0 + c$

$c = 0$

$t = \frac{1}{2} \tan^{-1} x$

$\therefore 2t = \tan^{-1} x \Rightarrow \boxed{x = \tan 2t}$ $\boxed{2}$

(111)

$$x = \tan 2t$$

$$v = 2 \sec^2 2t$$

↓ 1

$$dt \quad t = \frac{\pi}{8}$$

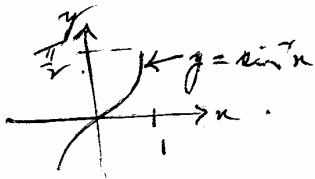
$$v = 2 \times \sec^2 \frac{\pi}{4}$$

$$= 2 \times (\sqrt{2})^2$$

$$= \boxed{4 \text{ m s}^{-1}}$$

2

(C)



$$v = \pi \int_0^1 [\sin^{-1} x]^2 dx$$

$$= \pi \times \frac{1}{3} \left[(\sin^{-1} 0)^2 + 4 \left[\sin^{-1} \frac{1}{2} \right]^2 + (\sin^{-1} 1)^2 \right]$$

$$= \frac{\pi}{6} \left[0^2 + 4 \times \left(\frac{\pi}{6} \right)^2 + \left(\frac{\pi}{2} \right)^2 \right]$$

$$= \frac{\pi}{6} \left[0 + \frac{\pi^2}{9} + \frac{\pi^2}{4} \right]$$

$$= \frac{\pi}{6} \times \frac{13\pi^2}{36}$$

$$= \frac{13\pi^3}{216}$$

$$= \boxed{1.87 \text{ m}^3 (\text{r.p.})}$$

3

QUESTION 6.

$$\begin{aligned} (a) \quad (i) \quad v^2 &= 28 + 24x - 4x^2 \\ &= 4(7 + 6x - x^2) \\ &= 4(7-x)(1+x). \end{aligned}$$

Clearly $v^2 \geq 0$

$$\therefore 4(7-x)(1+x) \geq 0$$

$$\therefore \boxed{-1 \leq x \leq 7}$$

1

$$(ii) \quad \text{Amplitude} = \frac{7 - (-1)}{2} = \boxed{4}$$

1

$$(iii) \quad \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d}{dx} (14 + 12x - 2x^2)$$

$$= 12 - 4x$$

$$= \boxed{-4(x-3)}$$

$$\left(\text{NB } v^2 = 4 \right. \\ \left. v = 2 \right)$$

$$(iv) \quad T = \frac{2\pi}{\omega} = \frac{2\pi}{2}$$

$$= \boxed{\pi \text{ secs}}$$

and centre of motion is $x=3$

$$(v) \quad x = 3 + 4 \cos(2t + \epsilon)$$

if $x=7$ when $t=0$.

$$7 = 3 + 4 \cos \epsilon$$

$$4 = 4 \cos \epsilon$$

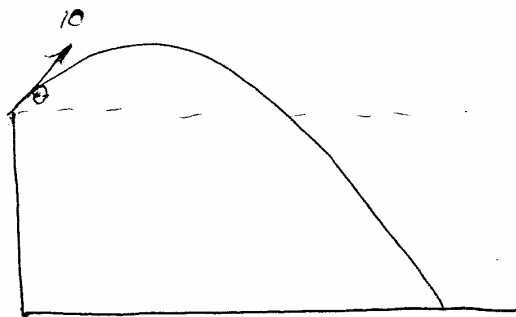
$$\cos \epsilon = 1$$

$$\epsilon = 0$$

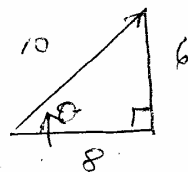
$$\therefore \boxed{x = 3 + 4 \cos 2t}$$

2

(b).



$$t=0, x=0, y=8$$



$$\text{NB } \theta = \tan^{-1}\left(\frac{6}{8}\right)$$

(1)

$$\ddot{x} = 0$$

$$\dot{x} = 8$$

$$x = 8t + C_1$$

$$\text{when } t=0, x=0 \therefore C_1=0$$

$$\boxed{x = 8t}$$

$$\ddot{y} = -g$$

$$\dot{y} = -gt + C_2$$

$$\text{clearly } \dot{y} = 6 \text{ when } t=0 \therefore C_2 = 6$$

$$\dot{y} = -gt + 6$$

$$\therefore y = -\frac{gt^2}{2} + 6t + C_3$$

$$\text{when } t=0, y=8 \therefore C_3=8$$

$$\therefore \boxed{y = -\frac{1}{2}gt^2 + 6t + 8}$$

3

(11). If $y=0$.

$$-5t^2 + 6t + 8 = 0 \Rightarrow -(5t^2 - 6t - 8) = 0$$

$$-(5t+4)(t-2) = 0$$

$$t = 2, -\frac{4}{5}$$

$\therefore$ 2 secs have elapsed.

1

and $\boxed{x = 16}$

1